CPT205 – Computer Graphics
Assessment 2 – 3D Modelling Project Report

Student Information
Name: Wu Xingjian
Student ID: 2361742
Programme: BSc Information and Computer Science
Project Title: Third-Person Futuristic City with Sci-Fi Elements
Date: December 2025

1. Introduction

This report presents the design and implementation of a 3D futuristic city scene developed for CPT205 – Computer Graphics Assessment 2. The primary objective of this project was to create an interactive, visually engaging 3D environment that demonstrates proficiency in key computer graphics concepts covered throughout the module, including hierarchical modelling, lighting, texture mapping, and animation.

The resulting application features a fully controllable third-person perspective within a dynamic sci-fi cityscape, complete with animated objects, day-night cycles, and interactive elements. This report will provide a comprehensive overview of the scene's design philosophy, the graphics techniques employed, and instructions for running the application.

2. Scene Design and Conceptual Framework

The central concept behind this project was to create a living, breathing futuristic city that evolves between day and night modes, offering distinct visual experiences in each setting. The city was designed with careful consideration of scale, movement, and atmospheric elements to create an immersive environment.

The city layout consists of an 11×11 grid of uniquely designed buildings surrounding a central public plaza. This plaza serves as the focal point for interactive elements, including the player-controlled robot and a hovering drone. The deliberate variation in building heights and architectural features creates visual diversity, while the inclusion of aerial elements such as a flying UFO and magnetic train adds dynamic movement to the skyline.

A key design principle was the contrast between day and night modes. During daytime, the scene features bright ambient lighting and clear visibility, while nighttime transforms the city with warm, glowing windows, illuminated street lights, and atmospheric lighting effects that create a distinctly different mood.

3. Implementation of Graphics Techniques

The project successfully implements a wide range of computer graphics techniques to achieve its visual goals:

Hierarchical Modelling and Animation

The player-controlled robot employs hierarchical modelling, with independently animated limbs that respond

to movement inputs. This structure allows for realistic walking animations where arms and legs swing in coordinated patterns. Similarly, the hovering drone features a hierarchical model with rotating blades attached to its main body.

Lighting and Material Systems
Two distinct lighting modes were implemented to support the day-night cycle. Day mode utilizes bright ambient and diffuse lighting to simulate sunlight, while night mode employs darker ambient settings complemented by localized light sources. Materials were carefully selected and applied to different objects, with particular attention to emission properties for glowing elements like building windows and vehicle thrusters.

Texture Mapping and Visual Detail
Custom texture mapping was implemented using a BMP loader, allowing for detailed surface treatments across the scene. Textures include building facades, road surfaces, ground cover, and specialized elements like the robot's facial features. The skybox system provides immersive environmental backgrounds that change between day and night modes.

Camera and Interaction Systems
A third-person follow camera was developed to provide smooth, intuitive navigation. The camera maintains a consistent offset behind the player character while responding to mouse input for rotation control. Interactive elements include keyboard-controlled movement, jumping mechanics with custom gravity logic implemented via kinematic equations, and the ability to toggle between day and night modes.

README

4. Program Execution and User Interaction

System Requirements and Compilation
The application requires a system with OpenGL compatibility and the FreeGLUT library. Compilation can be achieved using standard C++ compilers with appropriate OpenGL linking flags.

User Controls
The interface is designed for intuitive interaction:

- **W/A/S/D keys control robot movement**
- **Mouse movement adjusts the camera perspective**
- **Spacebar initiates jumping actions**
- **N key toggles between day and night modes**
- **ESC key exits the application**

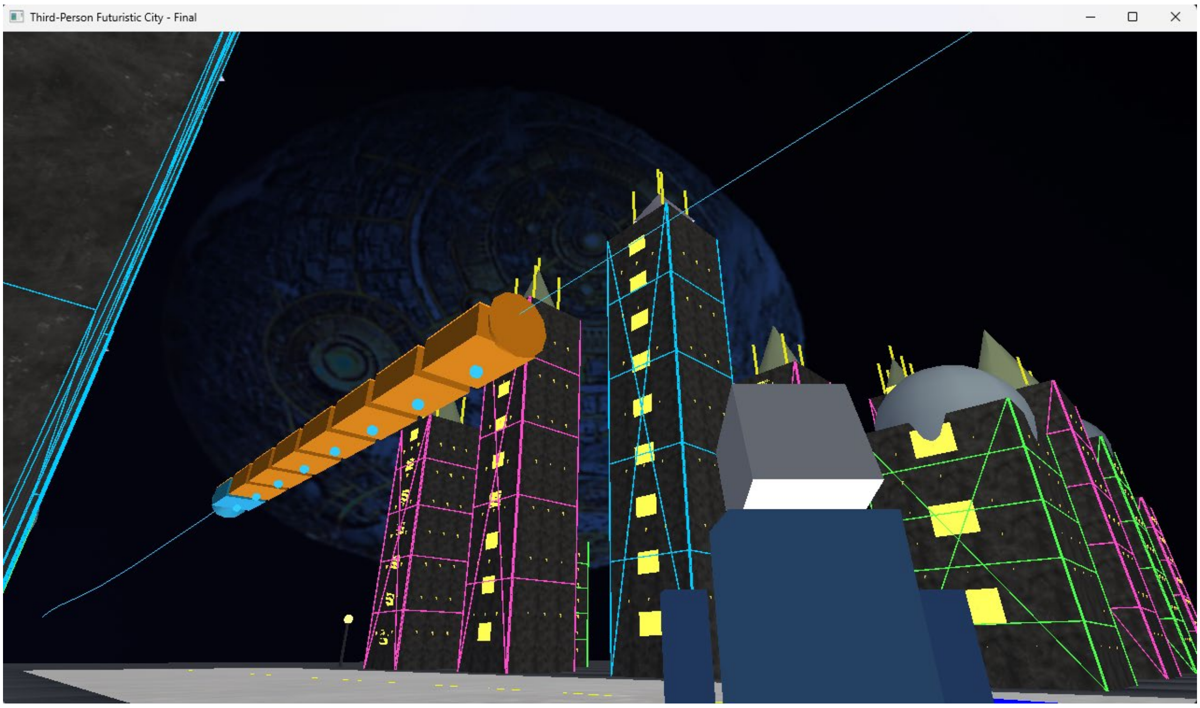
File Structure
The application requires several texture files in BMP format, including sky textures for both day and night modes, ground and road surfaces, building textures, and the robot's face texture. Please ensure the 'texture image files' folder is in the same directory as the executable for the BMP loader to work correctly.

5. Visual Documentation

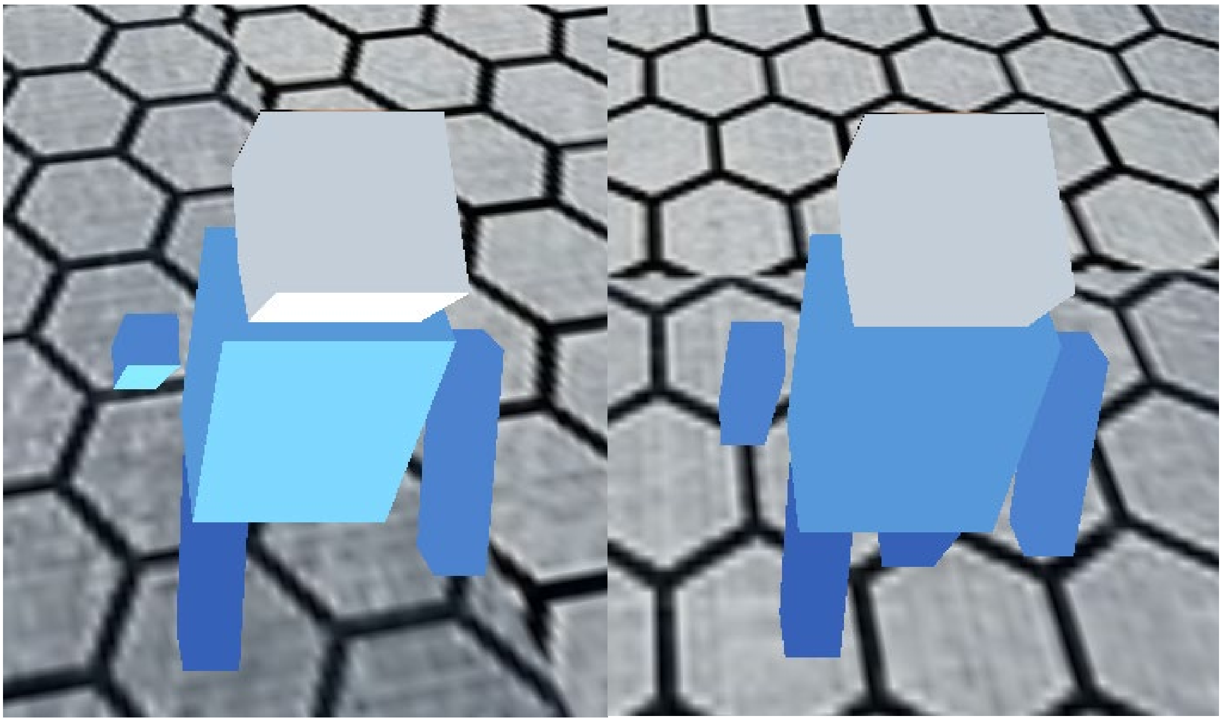
The application produces several distinctive visual scenarios worthy of documentation:

1. **Daytime Cityscape - Showcasing the full extent of the city layout with clear lighting and detailed building structures**

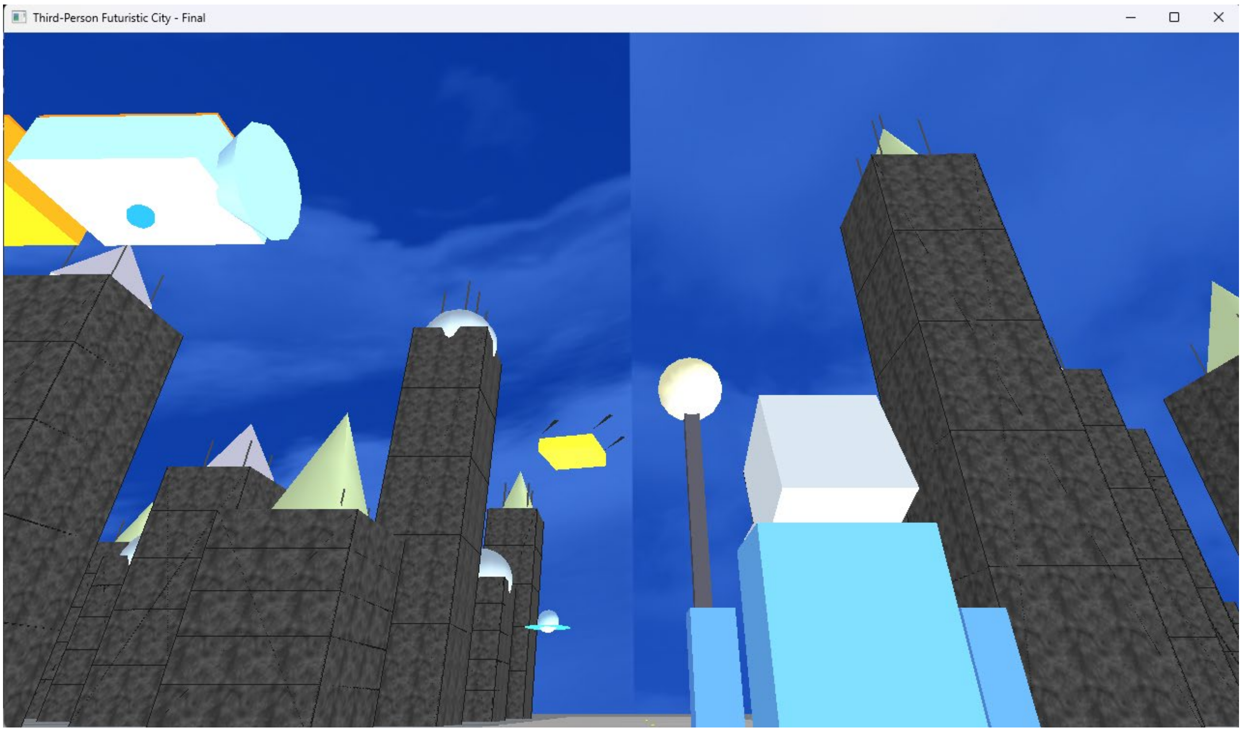
2. **Nighttime Transformation - Demonstrating the atmospheric lighting effects, glowing windows, and warm street illumination**



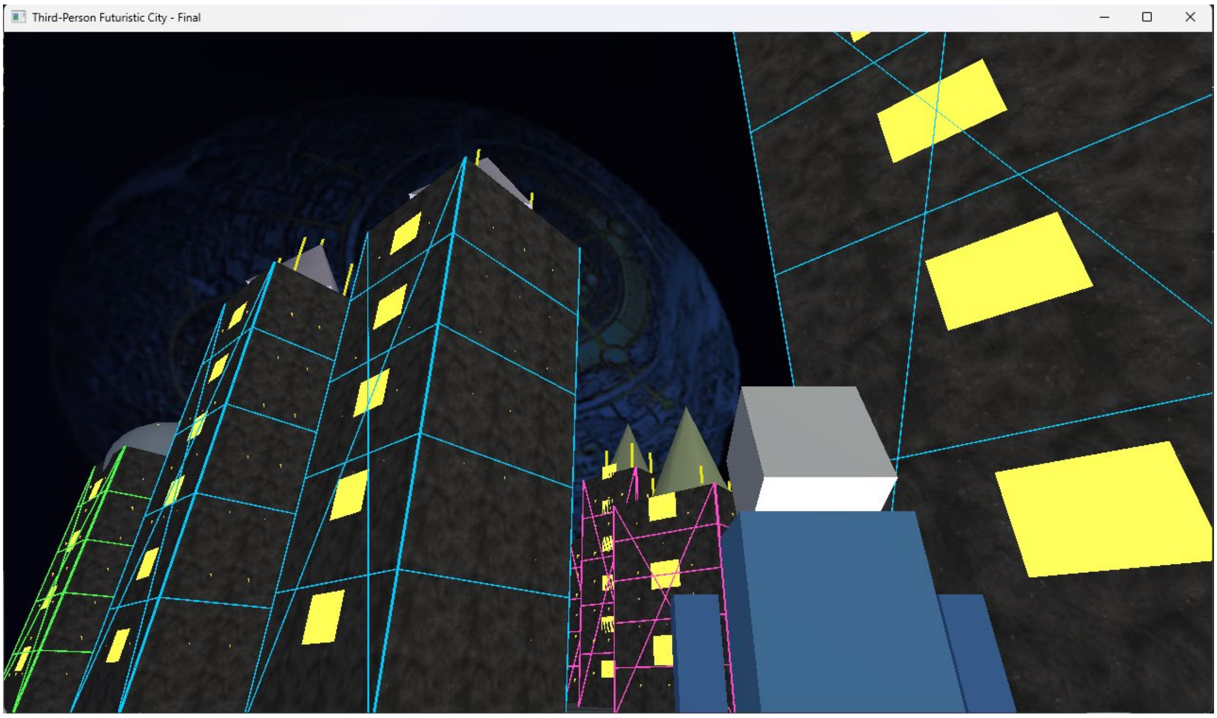
3. **Character Interaction - Illustrating the third-person perspective and robot animations within the central plaza**



4. **Aerial Elements - Capturing the UFO's circular flight pattern and the magnetic train's movement across the skyline**



5. **Architectural Details - Highlighting the variety of building designs and sci-fi decorative elements**



6. Conclusion

This project successfully demonstrates comprehensive understanding and application of computer graphics principles taught in CPT205. The futuristic city scene incorporates a wide range of techniques including hierarchical modelling, advanced lighting systems, texture mapping, and real-time animation.

References and Acknowledgements

Textures Assets

Environmental Textures: The textures applied to the building walls, ground, and road surfaces were sourced from TextureLabs (https://texturelabs.org/), a resource for free high-quality textures.

Robot Face: The facial texture for the robot character is a custom asset created from a personal photograph taken by the author.

Skybox (Day Mode): The daytime sky texture is an original photograph captured by the author.

Skybox (Night Mode): The nighttime sky texture is derived from a screenshot of the "Ecumenopolis" planet type from the video game *Stellaris* (Paradox Interactive). The image was subsequently processed and digitally modified by the author using Adobe Photoshop to create a seamless skybox suitable for the scene.